

# 1 Configurations for the Display of Actual and Target Cycle

## Purpose

In a production using clocks, the display of the current actual and target cycle of a machine is an important information. The system provides different options or units to display the cycle:

- Seconds/1000 clocks (is equal to milliseconds per 1 clock)
- Seconds/clock
- Clocks/minutes

This document describes the configuration options allowing the display of the actual and the target cycle on the shop floor terminal AIP 8.1 and 8.2 and on the MOC.

## Display of cycles on the MOC

The MOC shows the target/actual cycle in seconds per clock/stroke. You can display the cycle on the MOC in the following MOC applications:

- [Workplaces/Machines](#)
- [Cycle progression](#).



In the shop floor data collection (BDE), you can not only display the current actual cycle, but also the actual cycle of a specific operation. The calculation of the operation-related actual cycle is described in the document [OBJECT\\_MES-Operation\\_ActualData.pdf](#).

In the application *Configure syntactic types*, you can configure the cycle display on the MOC. This is described in the document [MOC\\_Configuration.pdf](#) in section "2 Change table data - Cycle configuration".



If communication to the server is not possible, the MOC continues to show the last value. This value is also shown if the machine is not in production any more.



The actual cycle displayed on the MOC can be different to the actual cycle displayed on the AIP 8.2 because the actual cycle is only sent to the server in case of specific events (dialogs M\_AST, M\_MST).

## Display of cycles on the terminal AIP 8.1

Via customization, you can display the actual cycle/target cycle in the dynamic dialogs on the AIP that show machine information (MMINFO and MINFO). The following formats are supported:

- Seconds/1000 clocks (is equal to milliseconds per 1 clock)

- Seconds/clock
- Clocks/minutes

To this end, you must store the acronym MNR.IZY or MNR.SZY in the relevant dynamic dialogs. In AIP standard processing, the field is automatically downloaded when data is reloaded.



If you use a customized version of the ctaiplay.ini file, this file could overwrite the downloaded field.

To display the respective format, you must store the required acronym in the dialogs:

| Meaning                                                    | Identifier in dynamic dialog<br>[target cycle] | Identifier in dynamic dialog<br>[actual cycle] |
|------------------------------------------------------------|------------------------------------------------|------------------------------------------------|
| Seconds/1000 clocks (is equal to milliseconds per 1 clock) | MNR.SZY                                        | MNR.IZY                                        |
| Seconds/clock                                              | MNR.SZY/HUB                                    | MNR.IZY/HUB                                    |
| Clocks/minutes                                             | MNR.SZY/MIN                                    | MNR.IZY/MIN                                    |

- Display "actual cycle [sec/cycle] / [sec/stroke]" using "MNR.IZY/HUB"
  - Acronym: **MNR.IZY/HUB** with name **Actual cycle [sec/cycle]** or **Actual cycle [sec/stroke]**
- Display "actual cycle [cycles/min.] / [strokes/min.]" using "MNR.IZY/MIN"
  - Acronym: **MNR.IZY/MIN** with name **Actual cycle [cycles/min.]** or **Actual cycle [strokes/min.]**



To display the target cycle in the relevant format, you must make the above acronym definition for SZY (see table above).

## Display of cycles on the terminal AIP 8.2

The document [AIP2 Configuration GUI.pdf](#) describes the configuration options that are available to display the actual cycle or target cycle in the tile view on the AIP 8.2. Refer to the sections *Constants (Defines)*, paragraph *FORMAT\_CYCLE*, and the section *CalculatedFields* to this end.

To display the actual cycle or the target cycle in the dynamic dialogs, e.g. in the machine information, the configuration options described in the previous section apply (*Display of cycles on the terminal AIP 8.1*).

## Sample configuration of dialog "Change target cycle (M\_SZY)"

### Field: Display Current target cycle

For the current target cycle, the displayed format is cycle/min in this example. The configuration must be set as follows:

#### Tab General:

| General        | Position                                                                                                             | Format | Functions | Options |
|----------------|----------------------------------------------------------------------------------------------------------------------|--------|-----------|---------|
| Activated      | <input checked="" type="radio"/> Always<br><input type="radio"/> Console only<br><input type="radio"/> Terminal only |        |           |         |
| Field no.      | 23                                                                                                                   |        |           |         |
| Unit           | [Sec/cycle]                                                                                                          |        |           |         |
| Information    | Current target cycle                                                                                                 |        |           |         |
| Identification |                                                                                                                      |        |           |         |
| ID index       |                                                                                                                      |        |           |         |

#### Tab Functions:

| General              | Position       | Format | Functions | Options |
|----------------------|----------------|--------|-----------|---------|
| Field attribute 1    | @MNR.SZY/MIN@  |        |           |         |
| Field attribute 2    | NOBORDER       |        |           |         |
| Field attribute 3    | FIELDLABELFONT |        |           |         |
| Field attribute 4    | STATUS         |        |           |         |
| Field attribute 5    |                |        |           |         |
| Field attribute 6    |                |        |           |         |
| Field attribute 7    |                |        |           |         |
| Field attribute 8    |                |        |           |         |
| Dialog list function |                |        |           |         |
| Combobox function    |                |        |           |         |
| Select file          |                |        |           |         |
| Function 1           |                |        |           |         |
| Function 2           |                |        |           |         |

### Field: Input field - New target cycle

For the new target cycle, the displayed or entered format is cycle/min in this example. The configuration must be set as follows:

### Tab General:

| General        | Position                                                                                                             | Format | Functions | Options |
|----------------|----------------------------------------------------------------------------------------------------------------------|--------|-----------|---------|
| Activated      | <input checked="" type="radio"/> Always<br><input type="radio"/> Console only<br><input type="radio"/> Terminal only |        |           |         |
| Field no.      | 29                                                                                                                   |        |           |         |
| Unit           | [Sec/cycle]                                                                                                          |        |           |         |
| Information    | New target cycle                                                                                                     |        |           |         |
| Identification | SZY                                                                                                                  |        |           |         |
| ID index       |                                                                                                                      |        |           |         |

### Tab Functions:

| General              | Position      | Format | Functions | Options |
|----------------------|---------------|--------|-----------|---------|
| Field attribute 1    | @MNR.SZY/MIN@ |        |           |         |
| Field attribute 2    | FOCUS         |        |           |         |
| Field attribute 3    | MANUELL       |        |           |         |
| Field attribute 4    |               |        |           |         |
| Field attribute 5    |               |        |           |         |
| Field attribute 6    |               |        |           |         |
| Field attribute 7    |               |        |           |         |
| Field attribute 8    |               |        |           |         |
| Dialog list function |               |        |           |         |
| Combobox function    |               |        |           |         |
| Select file          |               |        |           |         |
| Function 1           |               |        |           |         |
| Function 2           |               |        |           |         |

## Special feature in dialog *Change target cycle* (M\_SZY)

The dynamic dialog provides the display of the current actual and target cycle and a field to enter a new target cycle. To display the current values, you must change the display as described above.

To change the format of the input field, note the following:

You use the field with the identifier SZY:MODE to specify the compensation rule for the value entered. This field is not visible and can include the following values for the attribute *Default*:

- SZY/MIN cycles per minute
- SZY/MSEC milliseconds per cycle
- SZY/HUB seconds per cycle (default processing)



The new field for the compensation rule is already included in the dialog <M\_SZY / AIPDEF / 0 > with all new installations from SP13. If the field is not available, you must manually insert it in the dynamic dialog M\_SZY.

If you must manually insert the field in the dynamic dialog M\_SZY, the fields must be assigned as follows:

| Attribute         | Content  | Default or proposal |
|-------------------|----------|---------------------|
| Text              | SZY:MODE | Proposal            |
| Information       | SZY:MODE | Proposal            |
| Identifier        | SZY      | Default             |
| ID index          | MODE     | Default             |
| Default           | SZY/MIN  | Default             |
| Field attribute 1 | STATUS   | Default             |
| Visible           | -        | Proposal            |



Also change / extend and activate all dialogs that are specified for specific terminals or terminal groups!